CS 4400: Artificial Intelligence

Introduction

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What is AI?
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The Turing test
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What is Intelligence?

A wish-list of general characteristics of intelligence

- **Perception**: manipulation & interpretation of data provided by sensors
- **Action**: control and use of effectors to accomplish a variety of tasks
- **Reasoning**: deductive (logical) inference, inductive inference
- **Learning**: adapting behavior to better cope with changing environments, discovery of patterns, learning to reason, plan, and act.
- **Communication**: with other intelligent agents including humans using signals, signs, icons, …
- **Planning**: formulation of plans – sequences or agenda of actions to accomplish externally or internally determined goals

AI in pop culture - humanoids

C3PO

Star Wars (1977)

T-800

The Terminator (1984)

David / Gigolo Joe

A.I. (2001)

Sonny

I, Robot (2004)



AI in pop culture - systems

HAL 9000

2001: a Space Odyssey (1968)

Skynet

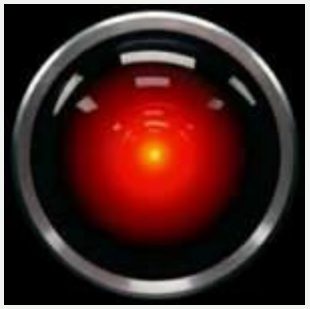
The Terminator (1984)

The matrix / agent Smith

The Matrix (1999)



AI in pop culture - systems



- HAL 9000

  2001: a Space Odyssey (1968)

  - part of the story centers around an intelligent computer called HAL
  - HAL is the "brains" of an intelligent spaceship
  - in the movie, HAL can
    - ✓ speak easily with the crew
    - ✓ see and understand the emotions of the crew
    - ✓ navigate the ship automatically
    - ✓ diagnose on-board problems
    - ✓ make life-and-death decisions
    - ✓ display emotions

  - In 1969 this was science fiction: is it still science fiction?
    https://www.youtube.com/watch?v=dKZczUDGp_I

Ethics

- People might lose jobs
- People might have too much leisure time
- People might lose sense of uniqueness
- People might lose privacy rights
- People might not be held accountable for certain actions
- Machines may replace the human race...



Different Types of Artificial Intelligence

- **Modeling exactly how humans actually think**
  - cognitive models of human reasoning
- **Modeling exactly how humans actually act**
  - models of human behavior (what they do, not how they think)
- **Modeling how ideal agents "should think"**
  - models of "rational" thought (formal logic)
  - note: humans are often not rational!
- **Modeling how ideal agents "should act"**
  - rational actions but not necessarily formal rational reasoning
  - i.e., more of a black-box/engineering approach
- **Modern AI focuses on the last definition**
  - we will also focus on this "engineering" approach
  - success is judged by how well the agent performs
  - modern methods are also inspired by cognitive & neuroscience (how people think).

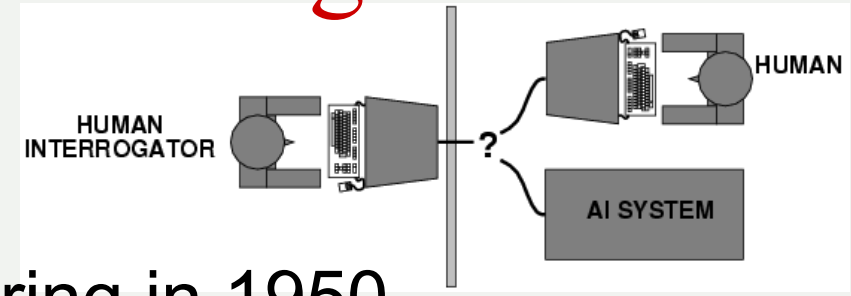
What is AI? Common definitions

System that thinks like humans	System that thinks rationally
System that acts like humans	System that acts rationally

"Thinking" vs. "Acting" (--)

"Human-like" vs "Rational" (|)

Acting Humanly: The Turing Test



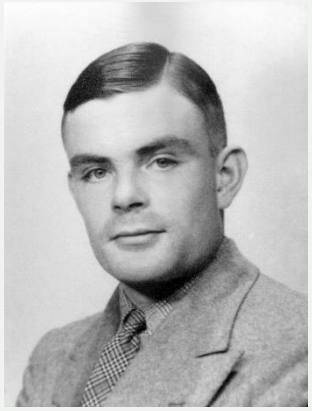
Test designed by Alan Turing in 1950

- A human interrogator (room1), two subjects (room 2)
  - Two subjects A and B: one human and one machine
- Using keyboards and screens, the interrogator asks written questions and receives written answers
- If interrogator cannot tell if A or B is human, then the machine passes
- No direct physical interaction between the interrogator and the machine
  - Physical simulation of a person unnecessary for intelligence

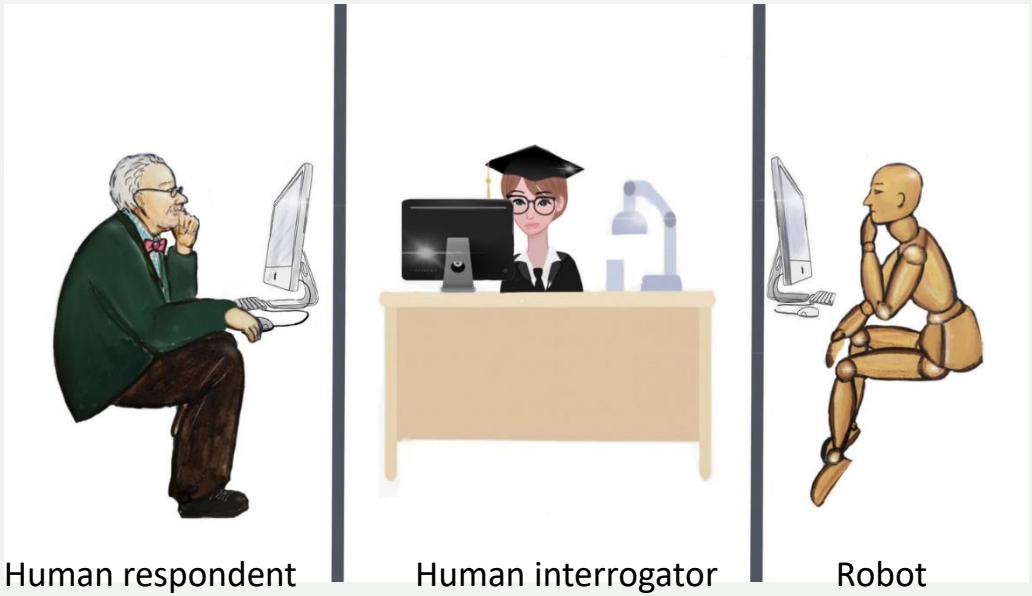
Acting Humanly: The Turing Test

Alan Turing (1950)

Operational definition of intelligence



https://www.britannica.com/biography/Alan-Turing



https://wsimag.com/science-and-technology/36961-no-turing-test-for-consciousne1ss

- A human interrogator poses some written questions to another human and a computer (or “robot”).
- The computer passes the test if the interrogator cannot tell whether the written responses come from the human responder or the computer.

Annual Loebner prize competition (since 1990): the first prize of $100,000 to be awarded to the first program that passes the "unrestricted" Turing test.

Acting Humanly: The Turing Test

What's needed to pass the Turing Test?

- Natural Language Processing
  - to communicate, say, in written English
- Knowledge Representation
  - to store what it knows or hears
- Automated Reasoning
  - to use the stored information to answer questions and draw new conclusions
- Machine learning (ML)
  - to adapt to new circumstances

The Total Turing Test

In total Turing test, physical interaction is assumed: interrogator can pass or show objects to subjects A & B

- Computer vision to perceive objects
- Robotics to manipulate objects

For a full physical simulation

- Speech recognition
- Speech synthesis

Six Disciplines of AI

To pass the Turing test, the following capabilities are required:

- Natural language processing (NLP) to communicate in a human language;
- Knowledge representation to store what it knows;
- Automated reasoning to answer questions and draw new conclusions;
- Machine learning (ML) to adapt to new circumstances and to detect patterns.

Total Turing test requires interaction with the real world.

- Computer vision (CV) & speech recognition to perceive the world;
- Robotics to move around in the world and to manipulate objects.

Most vibrant and in-demand areas in AI today: ML, CV, NLP, and Robotics.

Breaking news?

AI program “Eugene Goostman” passed the Turing test for world’s first time in June 2014

- A chatting program impersonating a 13-year-old Ukrainian boy, created by Vladimir Veselov et al. (https://www.youtube.com/watch?v=njmAUhUwKys)
- 33% of the 30 judges were convinced that the bot was human

Does it mean we have a working AI program?

- Emphasis on the outcome, not the underlying process of intelligence

Thinking Humanly: Cognitive Modeling

To say a program thinks like a human, we must know humans think first.

Learn about human thought in three ways:

- introspection – catching thoughts as they go by;
- psychological experiments – observing a person in action;
- brain imaging – observing brain activities.

General problem solver (GPS) by Newell & Simon (1961)

Also compares reasoning steps to those in the human solution.

Cognitive science constructs theories of the human mind by combining

- computer models from AI
- experimental techniques from psychology

Thinking Humanly: Cognitive Modeling

- Understanding the human thought process is the main goal of cognitive science
  - Psychological experiments + computational models of thinking
- Cognitive science is not AI
  - Algorithm that works on computers does not mean it's how human brain works
  - Successful AI applications need not base on cognitive modeling
  - The two fields have influenced each other

Thinking Rationally: "Laws of Thought"

Logic:

- Syllogism from Greek philosophers
  - "Socrates is a man: all men are mortal: therefore, Socrates is mortal."
- By 1965, programs exists that could solve any solvable logical problems *in principle*

Practical problems:

- Not everything can be formulated logically
- Large amounts of computation needed to do complete inference

Acting Rationally

Focus of this course

Compared with other approaches

- Acting/thinking humanly: “rationally” can be defined more concretely than “humanly”
- Thinking rationally: correct inference is part of acting rationally
- Act to achieve best expected outcome, with limited information and/or limited resource

Acting Rationally

This course is about designing rational agents.

Rational behavior: doing the right thing



expected to maximize goal achievement,
given the available information

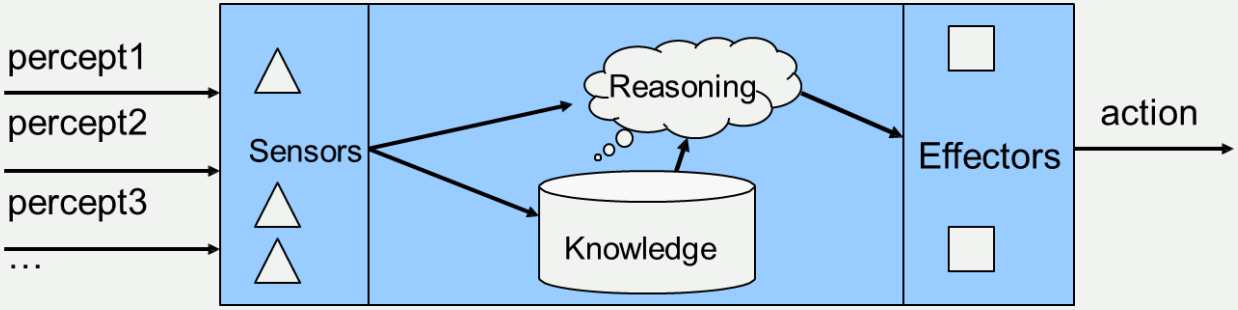
- An agent is an entity that perceives and acts.
- A rational agent is one that acts to achieve the best outcome.

Rational Agents (preview)

- **Rational behavior: Doing that is expected to maximize one's "utility function" in this world.**
- **An agent is an entity that perceives and acts.**
- **A rational agent perceives and acts rationally, requires knowledge representation and reasoning.**
- **This course is about designing rational agents**
- **Abstractly, an agent is a function from percept histories to actions:**

  [f: $\mathcal{P}^* \rightarrow \mathcal{A}$]
- **For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance**
- **Caveat: computational limitations make perfect rationality unachievable**

  → design best program for given machine resources



Limited Computation

Sometimes we don't have the resources

- Run out of time
- Run out of memory

Limited rationality: do the right thing given the constraints

Foundations of AI

□ **Philosophy**	**Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality.**
□ **Mathematics**	**Formal representation and proof, algorithms, computation, (un)decidability, (in)tractability, probability.**
□ **Economics**	**utility, decision theory, rational economic agents**
□ **Neuroscience**	**neurons as information processing units.**
□ **Psychology/ Cognitive Science**	**how do people behave, perceive, process information, represent knowledge, adaptation, phenomena of perception and motor control,**
□ **Computer engineering**	**building fast computers**
□ **Control theory**	**design systems that maximize an objective function over time**
□ **Linguistics**	**knowledge representation, grammar**

Landmarks in AI history

- 1943 McCulloh & Pitts: model of artificial neurons
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 McCarthy, Minsky, Newell, Simon, Shannon, Nash et al.
  Dartmouth workshop:"Artificial Intelligence" adopted
- 1952-69 Early enthusiasm, great expectations, optimism fueled by early success on some problems thought to be hard
- 1966-73 Collapse in AI research: Progress was slower than expected.
  Unrealistic predictions, Herbert Simon (1957)
  AI discovers computational complexity.

Landmarks in AI history

- 1969-86 Expert systems
- 1980- AI becomes an industry: expert systems booms, then busts (1988-93): “AI Winter”
- 1986- Neural networks regain popularity
- 1987- Probabilistic reasoning and machine learning

Landmarks in AI history

- 1995- Emergence of intelligent agents
  - AI technologies continue to find applications in
    - information retrieval
    - data mining and knowledge discovery
    - customizable software systems
    - smart devices (e.g., home, automobiles)
    - agile manufacturing systems
    - autonomous vehicles
    - bioinformatics
    - internet tools: search engines, recommender systems
    - …
  - Steady progress on fundamental AI research problems continues.

Landmarks in AI history

- 2001- Big data

  e.g., ImageNet

- 2011- Deep learning
  - Successful large-scale real-world applications in
    - image recognition
    - natural language processing
    - speech recognition
    - machine translation
    - …
  - Convolutional neural networks (CNNs)
  - Chat Generative Pre-trained Transformer (ChatGPT)

Turing Award Winners in AI

1969 Marvin Minsky (MIT)

1971 John McCarthy (Stanford)
National Medal of Science (1990)

1975 Allen Newell* (Carnegie Mellon)
National Medal of Science (1992)

Herbert Simon* (Carnegie Mellon)
Nobel Prize in Economics (1978)
National Medal of Science (1986)

Founding Fathers of AI

1994 Edward Feigenbaum (Stanford)
Raj Reddy (Carnegie Mellon)

2011 Judea Pearl (UCLA)

2018 Yoshua Bengio (U Montreal)
Geoffrey Hinton (U Toronto)
Yann Lecun (NYU & Facebook)

US News rankings of CS programs in AI (2024):

1. Carnegie Mellon
2. MIT
3. Berkeley
4. Stanford

…

*omitted from the textbook on p. 17. Big mistake!

Academic Genealogy

Solomon Lefschetz

Albert Tucker



Herbert Simon



John McCarthy



Marvin Minsky



Allen Newell



Edward Feigenbaum



Raj Reddy

* All the photos are from https://amturing.acm.org/byyear.cfm.

Current AI Applications

- Game playing
  - ♦ Chinook defeated human checkers champions (1994).
  - ♦ Deep Blue (IBM) defeated the reigning world chess champion Garry Kasparov (1997).
  - ♦ Supercomputer Watson (IBM) beat human champions on “Jeopardy” (2011).
  - ♦ AlphaGo (Google) beat the world’s No.1 ranked player Ke Jie in Go (2017).
  - ♦ Libratus (Carnegie Mellon) defeated the world’s best Texas Hold ‘em poker players head-to-head (2017), six-player (2019).
  - ♦ “AlphaStar” AI (Google) defeated human pros at StarCraft II (2019)

Current AI Applications

- Robotic vehicles
  - Stanley (Stanford) won the DARPA Grand Challenge (2005)
  - Boss (Carnegie Mellon) won the DARPA's Urban Challenge (2007)
  - Self-driving cars by Tesla, Google, etc.



Stanley



Boss

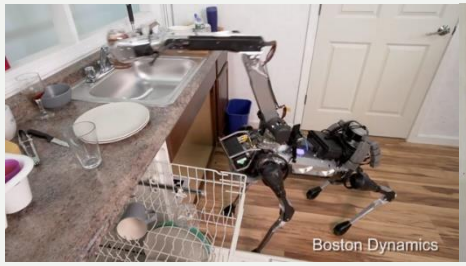


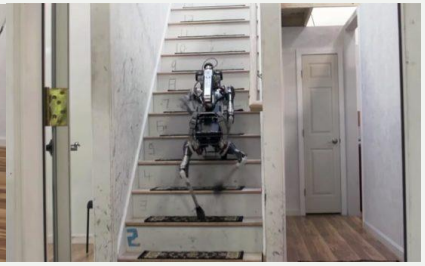
Google car



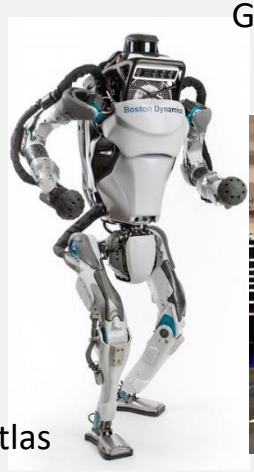
Tesla car crash
Florida, 2016

- Legged robots (Boston Dynamics)

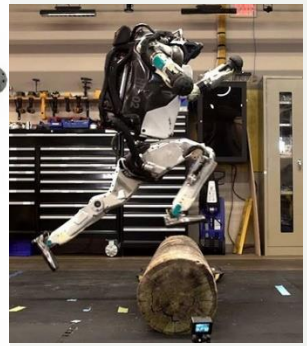




SpotMini



Atlas





Handle

(cont'd)

- Autonomous planning
  - NASA's Remote Agent program controlled the scheduling of operations for a space craft (2000).
  - Uber and Google Maps plan optimal routes for hundreds of millions of users.
- Speech recognition
  - Flight booking via conversation with an automated system
  - Real-time speech-to-speech translation
- Image understanding
  - ImageNet object recognition
  - Image captioning

  } Extensive use of convolutional neural networks (CNNs)
  - Face recognition (boarding airplanes, bullet trains, etc.)
- Natural language processing
  - ChatGPT by OpenAI allows user to steer and refine a conversation (2022)

Consider what might be involved in building a "intelligent" computer….

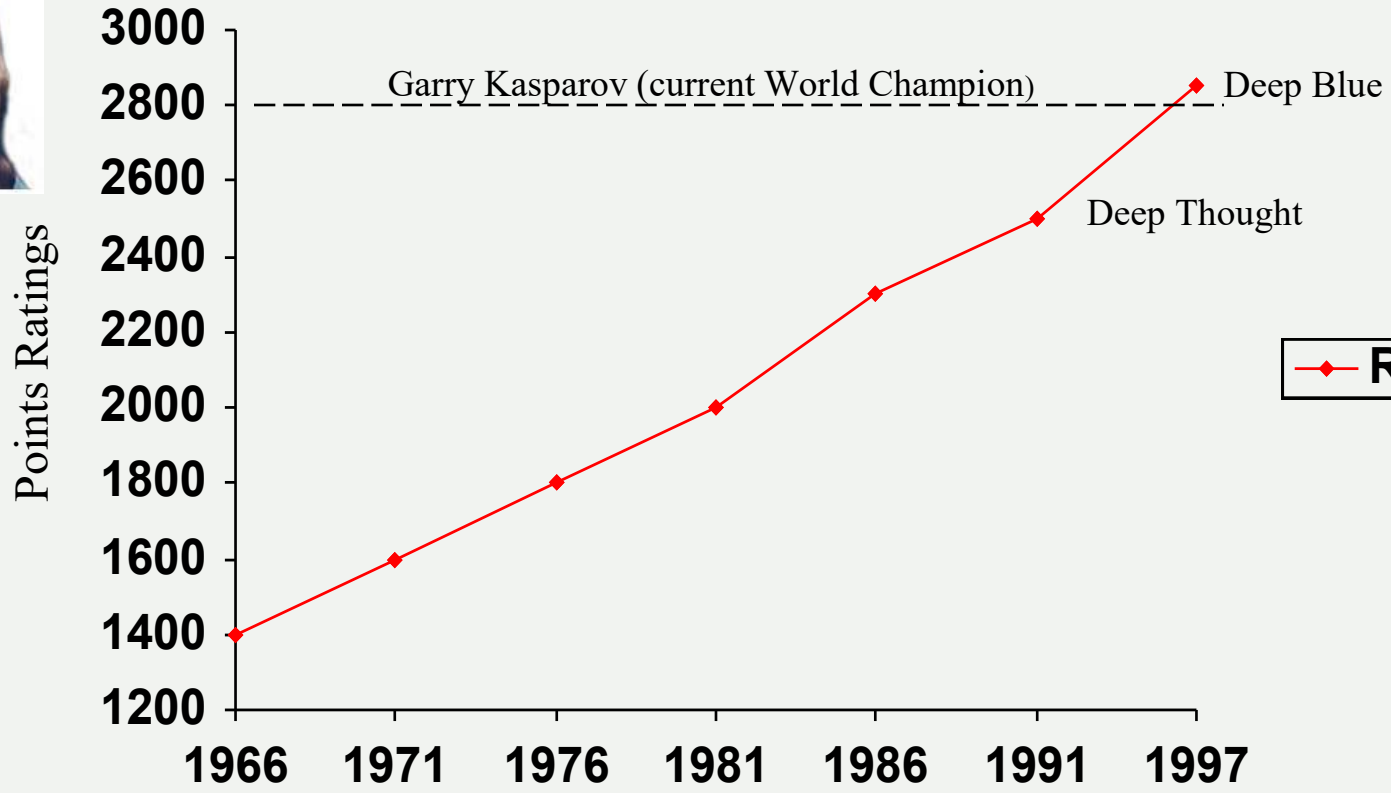
- **What are the "components" that might be useful?**
  - Fast hardware?
  - Foolproof software?
  - Speech interaction?
    - speech synthesis
    - speech recognition
    - speech understanding
  - Image recognition and understanding ?
  - Learning?
  - Planning and decision-making?

Can Computers play Humans at Chess?

- **Chess Playing is a classic AI problem**
  - well-defined problem
  - very complex: difficult for humans to play well







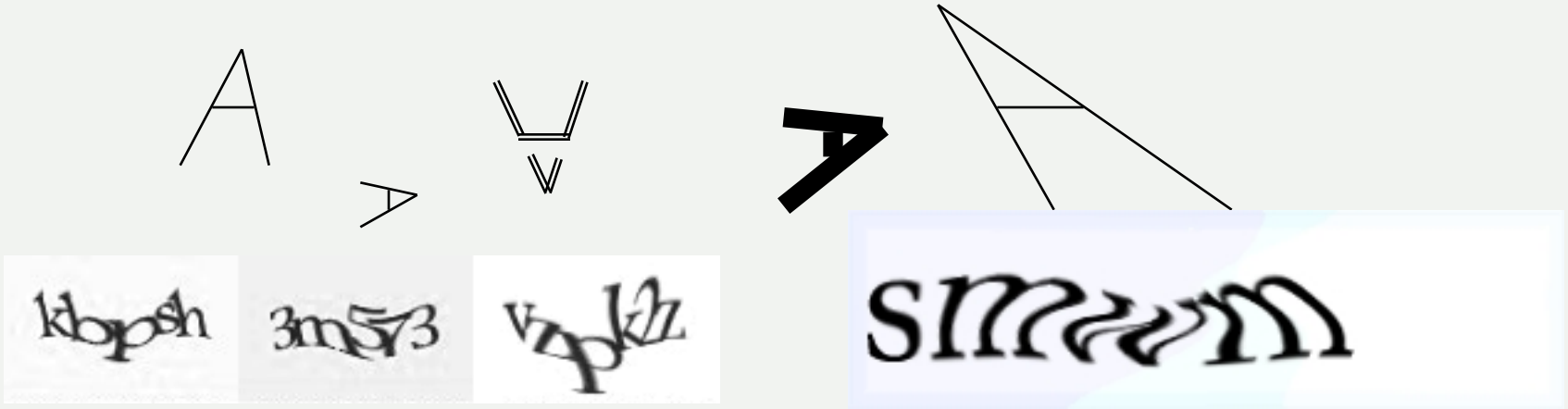
- **Conclusion: YES:** today's computers can beat even the best human

Can we build hardware as complex as the brain?

- **How complicated is our brain?**
  - a neuron, or nerve cell, is the basic information processing unit
  - estimated to be on the order of 10^{11} neurons in a human brain
  - many more synapses (10^{14}) connecting these neurons
  - cycle time: 10^{-3} seconds (1 millisecond)
- **How complex can we make computers?**
  - 10^6 or more transistors per CPU
  - supercomputer: hundreds of CPUs, 10^9 bits of RAM
  - cycle times: order of 10^{-8} seconds
- **Conclusion**
  - **YES**: in the near future we can have computers with as many basic processing elements as our brain, but with
    - far fewer interconnections (wires or synapses) than the brain
    - much faster updates than the brain
  - **but** building hardware is very different from making a computer behave like a brain!

Can Computers "see"?

- **Recognition v. Understanding (like Speech)**
  - Recognition and Understanding of Objects in a scene
    - look around this room
    - you can effortlessly recognize objects
    - human brain can map 2d visual image to 3d "map"
- **Why is visual recognition a hard problem?**



- **Conclusion: mostly NO:** computers can only "see" certain types of objects under limited circumstances: but **YES** for certain constrained problems (e.g., face recognition)

Can AI …

play a decent game of table tennis?

drive in Victorville, California?

drive in the center of Cairo, Egypt?

buy a week's worth of groceries at the market?

buying a week's worth of groceries on the Web?

play a decent game of bridge at a competition level?

Can AI …

discover and prove new mathematical theorems?

write an intentionally funny story?

give competent legal advice in a specialized area of law?

translate spoken English into spoken Swedish in real time?

perform a complex surgical operations?

Conclusion

- AI is about building intelligent agents (robots)
- There are many very interesting sub-problems to solve:
  - Learning, vision, speech, planning, …
- Surprising progress has been made (autonomous cars, chess computers) but surprising lack of progress is also a fact (visual object recognition).
- There is no doubts that AI has a bright future: technology is increasingly getting smart.

